Abbreviations

cao	correct answer only
dep	dependent
FT	follow through after error
isw	ignore subsequent working
oe	or equivalent
SC	Special Case
nfww	not from wrong working
soi	seen or implied

Question	Answer	Marks	Partial Marks
1(a)(i)	14, 10	**2**	**M1** for 24 ÷ (7 + 5)
1(a)(ii)	$\frac{3}{350}$	**2**	**B1** for correct fraction not in lowest terms
1(a)(iii)	120	**1**	
1(b)(i)	10.2[0]	**2**	**M1** for $\frac{15}{100}\times 12$ oe or better
1(b)(ii)	45	**2**	**M1** for $\frac{38.25}{1-\frac{15}{100}}$ oe
1(c)(i)	85	**2**	**M1** for $\frac{500\times 1.7\times 10}{100}$ oe
1(c)(ii)	203 or 202.5 to 202.6	**2**	**M1** for $200\times\left(1+\frac{0.0035}{100}\right)^{365}$
1(c)(iii)	1.9	**3**	**M2** for $\sqrt[6]{\frac{559.78}{500}}$ or **M1** for $500\left(1+\frac{r}{100}\right)^{6}=559.78$
2(a)(i)	$\begin{pmatrix}6\\17\end{pmatrix}$	**2**	**B1** for each
2(a)(ii)	6.4[0] or 6.403...	**2**	**M1** for $4^2 + 5^2$
2(b)	(1, 2)	**1**	
2(c)	(0, –2)	**1**	

Question	Answer	Marks	Partial Marks
2(d)	$\frac{1}{2}\mathbf{c} + \frac{1}{3}\mathbf{d}$	**3**	**B2** for correct unsimplified answer or **M1** for $\overrightarrow{CT} = -\mathbf{c} + \frac{2}{3}\mathbf{d}$ oe or $\overrightarrow{TC} = \mathbf{c} - \frac{2}{3}\mathbf{d}$ oe or for correct route
3(a)	41.4	**4**	**M1** for 10, 30, 42.5, 47.5, 55, 70 **M1** for Σfx where x lies in or on the boundary of each interval. **M1 dep** for $\frac{\Sigma fx}{200}$ dep on second M1
3(b)(i)	112, 170	**1**	
3(b)(ii)	Correct diagram	**3**	**B1** for correct horizontal plot **B1FT** for correct vertical plots **B1 FT** dep on at least B1 earned for reasonable increasing curve or polygon through their 6 points If 0 scored **SC1FT** for 5 out of 6 points plotted correctly
3(b)(iii)(a)	48	**1**	
3(b)(iii)(b)	160	**2**	**M1** for 40 seen
3(c)	$\frac{87}{3980}$ oe	**2**	**M1** for $\frac{30}{200} \times \frac{29}{199}$ oe
3(d)	Correct histogram	**3**	**B1** for each column If 0 scored **SC1** for correct frequency densities soi 1.25, 12, 1
4(a)	65.4 or 65.36 to 65.37	**3**	**M1** for $150^2 + 120^2 - 2 \times 150 \times 120 \cos 25$ **A1** for 4270 or 4272 to 4273
4(b)	125 or 124.7 to 124.8	**4**	**B1** for [angle S =] 80 **M2** for $\frac{150 \sin 55}{\sin \textit{their} 80}$ or **M1** for $\frac{\sin \textit{their} 80}{150} = \frac{\sin 55}{RS}$ oe
4(c)	10 400 or 10 410 to 10 440 nfww	**3**	**M1** for $\frac{1}{2} \times 120 \times 150 \sin 25$ oe **M1** for $\frac{1}{2} \times 150 \times \textit{their}$ **(b)** $\sin 45$ oe